

GREEN INDUSTRIES

Green industries courses offer students opportunities to investigate how to care for and sustainably manage our natural and living resources. Students will explore a variety of areas within green industries, including agriculture, floristry, forestry, horticulture, and/or landscaping.

Students will gain practical skills and knowledge by completing a variety of authentic, industry-relevant activities and projects, such as plant propagation and greenhouse maintenance activities; landscape design and planning exercises; assignments that include farm or forestry management plans; and various construction projects. In addition, students will be introduced to concepts pertaining to biodiversity, environmental sustainability, and natural versus artificial products. Students will also investigate the numerous and varied ways in which green industries and society are interdependent.

Green industries courses will prepare students for working safely in the many fields that are encompassed in this subject area. Students will learn about career opportunities in the green industries and about the Essential Skills and work habits that are important for success in these fields.

The list of approved emphasis areas for green industries can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Green Industries, Grade 11

Workplace Preparation

THJ3E

This course enables students to develop knowledge and skills related to agriculture, floristry, forestry, horticulture, and landscaping. Students will learn to identify a broad range of plant and animal species; examine factors that affect the growth of plants and animals and the quality of products derived from them; and develop process, design, and maintenance skills required in the green industries. Students will also learn about safe and healthy working practices, develop an awareness of environmental and societal issues related to green industry activities, and learn about apprenticeships and other postsecondary education and training opportunities, as well as employment opportunities that may be pursued directly after graduation.

Prerequisite: None

A. GREEN INDUSTRY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of species classification and identification and of relationships between species and geographical regions;
- A2.** demonstrate an understanding of the effects of biotic and abiotic factors on growth and product quality;
- A3.** demonstrate an understanding of design and planning processes and their application to a variety of requirements in the green industries;
- A4.** use mathematical, documentation, research, and communication skills as they apply to the green industries.

SPECIFIC EXPECTATIONS

A1. Species Classification and Geographical Regions

By the end of this course, students will:

- A1.1** distinguish between different plant and/or animal groups and identify them by key characteristics and desirable features (*e.g., annuals and perennials; native and non-native plants; major types, species, and varieties of trees, shrubs, flowering plants, and crops; animal breeds*);
- A1.2** identify geographical regions on the basis of classification criteria relevant to the green industries (*e.g., forest type, hardiness, soil type*);
- A1.3** explain the relationships between geographical regions and the key characteristics and desirable features of plant and/or animal groups within them (*e.g., relationship of plant and animal characteristics to available heat, moisture, light, shelter, and food*).

A2. Factors Affecting Growth and Product Quality

By the end of this course, students will:

- A2.1** identify the main abiotic factors that affect growth and post-harvest quality (*e.g., temperature, sunlight, soil composition, rainfall and soil moisture*);
- A2.2** describe biological processes that are essential to the propagation, development, and health of plants and/or animals and the quality of products derived from them (*e.g., photosynthesis,*

respiration, reproduction, transpiration, post-harvest physiology, digestion);

- A2.3** identify a variety of pests and diseases (*e.g., bacteria, viruses, moulds, fungi, insects, animals*) that may affect the health of plants and/or animals and the quality of products derived from them.

A3. Designs and Processes

By the end of this course, students will:

- A3.1** describe the steps in a design or planning process (see pp. 22–23) and demonstrate an understanding of their application to a variety of requirements in the green industries (*e.g., preparing environmental farm plans, urban forestry management plans, landscape designs; designing water gardens, mass arrangements*);
- A3.2** describe common operational processes that are used in the green industries (*e.g., single animal management, crop location and rotation, crop scheduling, event planning, waste management, composting, select cutting*);
- A3.3** identify a variety of structures used in the green industries (*e.g., cold frames, greenhouses, toolsheds, animal pens, storage facilities*), and describe how their structural features relate to their functions;
- A3.4** demonstrate an understanding of correct procedures for the care and handling of plants and/or animals (*e.g., propagating, pruning, transporting, watering, feeding, fertilizing*).

A4. Technological and Mathematical Literacy and Communication Skills

By the end of this course, students will:

A4.1 demonstrate an understanding of terminology used in the green industries and use it correctly in oral and written communication (*e.g., pulp, deciduous, form flower, scale, loam*);

A4.2 use effective documentation practices to record and track important information related to green industry operations (*e.g., preparing estimates, inventories, and invoices; maintaining cleaning and maintenance records, food source records, pesticide field logs, crop management records*);

A4.3 identify technical issues of current significance in the green industries, using information from general media sources (*e.g., newspapers, magazines, the Internet*);

A4.4 use appropriate calculations and units of measurement when completing a variety of green industry tasks (*e.g., calculating fertilizer and preservative applications, yields, basal areas; surveying; calibrating sprayers; converting between metric and imperial units*).

B. GREEN INDUSTRY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate an understanding of and apply design and production practices that are commonly used in the green industries;
- B2.** apply management strategies for assessing and controlling biotic and abiotic factors that affect plant and/or animal quality;
- B3.** demonstrate competence in technical skills relating to specific applications and tasks in the green industries.

SPECIFIC EXPECTATIONS

B1. Design and Production

By the end of this course, students will:

- B1.1** implement a production process or procedures according to a design or plan (*e.g., harvest a crop, construct a landscape, grow and cultivate plants, make floral arrangements*);
- B1.2** demonstrate an understanding of and apply techniques for the propagation and care of plants and animals and for ensuring the quality of products derived from them (*e.g., plant/tree regeneration, animal reproduction, cut flower processing, crop production, tree planting*);
- B1.3** demonstrate an understanding of and apply techniques and processes that promote biodiversity, increase ecosystem function, and reduce maintenance requirements (*e.g., planting of native species, mulching, naturalizing gardens, using local cut flowers*);
- B1.4** demonstrate an understanding of and apply marketing techniques that are commonly used in the green industries (*e.g., creation of product displays, posters, and pamphlets*).

B2. Plant and Animal Management Strategies

By the end of this course, students will:

- B2.1** apply a variety of methods to monitor biotic factors that affect plant and/or animal quality (*e.g., pest scouting, regular health inspections of animals, weed identification, post-harvest tracking of product freshness and quality*);

- B2.2** apply a variety of methods to monitor abiotic factors that affect plant and/or animal quality (*e.g., nutrient balancing, soil testing, monitoring indoor and outdoor environmental conditions*);

- B2.3** apply a variety of pest and disease control techniques (*e.g., crop rotation, greenhouse sanitation, enhancement of natural barriers, disinfection of equipment*).

B3. Technical Skills

By the end of this course, students will:

- B3.1** demonstrate competence in the basic technical skills required to complete a variety of tasks in the green industries (*e.g., rigging loads, operating power tools, wiring and taping flowers, using a guillotine cutter to cut natural and manufactured products, applying pesticide*);
- B3.2** construct a variety of structures or products that are used or produced in the green industries (*e.g., planter boxes, scaffolding, seasonal crafts, pavers, fences, skid trails*);
- B3.3** demonstrate competence in related technical skills (*e.g., welding, small-engine maintenance, use of computer applications, use of GPS equipment*) that are required to complete a variety of green industry projects.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** identify the impact of the green industries on the environment and describe ways of minimizing harmful effects;
- C2.** describe social and economic relationships and issues involving the green industries.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** describe the effects of green industry activities on the environment in the past and in the present (*e.g., destruction of habitat, increased energy use for long-distance shipping of floral products, pesticide and fertilizer contamination, greenhouse gas emissions from tillage and sheep and cattle, noise and air pollution from gasoline- and diesel-powered machinery*);
- C1.2** describe the advantages and disadvantages of using natural rather than manufactured materials or products in the green industries (*e.g., natural fertilizers and pest control methods rather than chemical fertilizers and pesticides, real flowers rather than artificial flowers, real grass rather than artificial turf, untreated rather than pressure-treated lumber*);
- C1.3** identify sustainable practices and guidelines that are currently applied within the green industries or may be applied in the future (*e.g., environmental farm planning, sustainable forest management, integrated pest management, sustainable golf course maintenance, select spraying, energy-efficient greenhouse production*);
- C1.4** describe the environmental implications (*e.g., effects on landfill lifespan and water and air quality*) of using particular materials, products, processes, and disposal methods (*e.g., chemically treated wood products; recycling, reusing, composting; using correct disposal methods for invasive plants*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** identify linkages between local communities and economies and the green industries (*e.g., consumer needs and green industry goods and services, green industries as sources of jobs and tax revenues, communities as providers of services to green industries and as regulators of green industry activities*);
- C2.2** describe societal issues relating to the green industries and identify ways of resolving them (*e.g., effects on Aboriginal hunting and harvesting territories, land use conflicts such as parkland versus commercial development, property rights and municipal landscape management, animal welfare, rights of migrant workers, fair trade concerns relating to imported agricultural or floral products*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and apply safe working practices as they relate to the green industries;
- D2.** identify careers in the green industries, and describe the skills, education, and training required for entry into these occupations.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify the personal protective clothing and equipment needed to perform various green industry tasks safely, and use as required to ensure their own and others' safety in the work environment (*e.g., eye and ear protection, hand and foot protection, head protection, sun protection, equipment guards*);
- D1.2** demonstrate an understanding of environmental and site-related hazards (*e.g., land conditions, weather conditions, dangerous plants and animals, utility lines, glass structures, hanging limbs, chicots*) and apply appropriate safety measures for avoiding them (*e.g., roping off danger areas, removing hazards, setting up traffic controls*);
- D1.3** demonstrate an understanding of and apply safe procedures for using and maintaining materials, tools, and equipment (*e.g., avoid moving parts and pinch points; perform a circle check of vehicles and equipment; check condition of materials, hydraulic systems, and protective equipment; check oil and fuel levels*);
- D1.4** demonstrate an understanding of and apply safe procedures for handling plants and/or animals (*e.g., dethorning plants, using ergonomic lifting techniques or devices, using hand protection, securing loads correctly for transport, understanding animal perception, avoiding actions that startle animals, using chutes and restraining devices*);
- D1.5** identify potentially hazardous situations in the workplace by conducting and documenting personal and workplace safety audits;

D1.6 identify sources of information about workplace hazards and how to avoid them (*e.g., Workplace Hazardous Materials Information System [WHMIS], Passport to Safety*);

D1.7 outline and comply with legislation for protecting the health and safety of workers in the green industries (*e.g., Occupational Health and Safety Act, local by-laws, fire prevention regulations*).

D2. Career Opportunities

By the end of this course, students will:

- D2.1** describe careers (*e.g., arborist, florist, herder, greenhouse worker, forester*) in the sectors of the green industries and the education, training, and certification required for entry into these occupations;
- D2.2** identify ways of acquiring knowledge and experience in green industry occupations (*e.g., through part-time work experience, cooperative education, guest speakers, field trips, job shadowing*);
- D2.3** identify the size of the labour force and the employment opportunities available in the green industries in their local community;
- D2.4** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the green industries (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);

D2.5 demonstrate an understanding of and apply the Essential Skills that are important for success in the green industries, as identified in the Ontario Skills Passport (*e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);

D2.6 demonstrate an understanding of and apply the work habits that are important for success in the green industries (*e.g., working safely, teamwork, reliability, initiative, customer service, and entrepreneurship*);

D2.7 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in areas related to the green industries (*e.g., Passport to Safety certificate, reports, assignments, Ontario Skills Passport Work Plan, industry certifications*), and explain why having a current portfolio is important for career development and advancement.